BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Trace Tables

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does a diamond represent in a flowchart?

2. In OCR ERL, which value can random(1,6) return?

A) 0 B) 7
C) 6 D) 1.5

3. "A USB flash drive stores data on a rotating magnetic platter." Correct this. Explain why.

4. A large file must cross a slow link. "Compression always makes files larger." Correct this. How might compression affect transfer time?

Today's objectives

Recall the essential ideas of trace tables. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Trace table
Records program variables and values as instructions execute.

Accumulator
Variable maintaining a running total.

Iteration counter
Variable tracking loop repetitions.

Assignment
Instruction replacing a variable's stored value.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Read statements in order	A trace records program state after each instruction that changes a variable.
Track an accumulator	An accumulator holds a running total; it changes each time a value is added.
Include selection and loops	Trace iteration counters, condition outcomes and updated values; do not record an update when a branch is skipped.
Check final output	The final displayed result should agree with the last recorded relevant value.

I Do - One worked example

Trace total = 0; for i = 1 to 4; total = total + i; next i; print(total). [5 marks]

1. Initialise variables

2. Before first iteration, total = 0.

3. After i=1,2,3,4 the totals are 1, 3, 6, 10.

4. The output is 10; rows record each update to total. [5]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Trace total=0; for i=1 to 3; total=total+2; next i. Give each value. [4]

2. Trace x=8; if x>5 then x=x-3 endif; print(x). [3]

3. Explain why trace tables can expose logic errors in a loop. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the loop statement that adds every file size.

```
array files = [4,6,10]
total = 0
for i = 0 to 2
...
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Trace x=2; x=x*3; x=x-1; print(x). Give x after each update and the output. [3]

Q3. Trace total=0; for i=1 to 3; total=total+i; next i. Record three totals. [3]

Q4. Explain why a trace must record a variable only after the relevant assignment has executed. [2]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table can help check how values change during a program. ☐ T ☐ F
2. A trace records program state after each instruction that changes a variable. ☐ T ☐ F
3. An accumulator holds a running total; it changes each time a value is added. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A decision or condition with alternative paths. ☐

Q2 C - 6 ☐

Q3 Solid-state storage with no moving read/write mechanism. ☐

Q4 A smaller file uses less storage and can be sent faster over the same link. Compression and decompression take processing time. ☐

Code First - Model

```
array files = [4,6,10]
total = 0
for i = 0 to 2
total = total + files[i]
next i
print(total)
```

Output: 20

We Do - Check the model

1. After i=1 total 2 [1], i=2 total 4 [1], i=3 total 6 [1], final output 6 [1]. ☐

2. Condition 8>5 is true [1]; x becomes 5 [1]; output 5 [1]. ☐

3. They show values after each iteration [1], letting a programmer see where an unexpected value appears [1], compare with intended behaviour [1], and locate the faulty update or condition [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 After multiplication x=6 [1]; after subtraction x=5 [1]; output 5 [1]. ☐

Q3 1 [1], 3 [1], 6 [1]. ☐

Q4 Before the assignment the old value is held [1]; recording afterwards shows the new program state correctly [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: